3Consulting
BUILD TRUST IN THE DIGITAL ECONOMY

FRAUD PREVENTION · CASE STUDY

# FraudGuard: Risk Scoring & Analyst Queue

BI Scientist • AI/ML Engineer • Python Developer

Fraud scoring and analyst platform: Kafka pipeline, XGBoost + Isolation Forest, 0–1000 risk score and SAR workflow. Deployment pending verification.

PAYMENT VULNERABILITIES

# The Speed & Precision Dilemma in Modern Payments

Payment and commerce operators must stop fraud without blocking genuine customers or burying analysts in alerts.

## Account Takeover (ATO)

- Credential stuffing and SIM-swap fraud breach accounts in seconds.
- Sudden IP, device fingerprint, and geographic jumps bypass static rules.
- Need dynamic velocity counters across 1-min to 24-hr windows.

## Card Velocity Bursts

- Fraud rings test stolen BIN ranges with micro-charges.
- Standard daily volume limits fail to detect rapid 10-second bursts.
- Leads to chargebacks and processor penalties.

## Analyst Fatigue & SARs

- Static rules flood analysts with false-positive alerts.
- Legitimate payments get blocked while real fraud slips through.
- Regulatory Suspicious Activity Reports (SARs) are delayed.

SYSTEM TOPOLOGY

# Transactional Outbox, Kafka & Ensemble Scoring

Each prediction is written once, by the streaming consumer, with reasons for the analyst.

## Ingestion

- Transactions land in PostgreSQL.
- Outbox relay publishes them to Kafka.
- Consistent across service restarts.

## Dynamic Risk Engine

- Shared feature definitions for training and scoring.
- XGBoost + Isolation Forest ensemble, 0–1000 score.
- Allow, review or block bands.

## Analyst Workbench

- Prioritised review queue and case notes.
- SAR triage and audit trail.
- Analyst labels feed scheduled retraining.

ANALYST WORKBENCH

# FraudGuard Operations Console

Analyst console for scored transactions and risk reasons (demo data).

- **Score Bands:** Low-risk activity passes, uncertain cases go to an analyst, high-risk activity is blocked or escalated.
- **Risk Reasons:** Plain-language reasons and shared device, IP and phone links shown with each case.
- **SAR Workflow:** Confirmed cases move into a Suspicious Activity Report triage queue with an audit trail.

Kafka | XGBoost | PostgreSQL | Airflow + MLflow




fraudguard.3consult-ng.com/app/queue

WHERE IT STANDS

# Built, Pilot Preparation

What exists in the codebase today. No customer or performance results claimed.

Status
**Pilot prep**
Deployment pending verification

Score
**0–1000**
With risk reasons per case

Models
**XGB + IF**
Calibrated ensemble, MLflow-tracked

Stack
**Compose**
Kafka, Postgres, Airflow, Grafana

## Operational Efficiency

- Review queue is ordered by risk, so analysts see the riskiest cases first.
- Analyst decisions are stored as labels for retraining.
- Audit trail on every case decision.

## Scalable Infrastructure

- Airflow DAGs for retraining, promotion, drift and bias checks.
- Prometheus, Grafana and Alertmanager for monitoring.
- Needs a Docker host; not shared hosting.

CASE STUDY TAKEAWAY

# Score every transaction. Show the analyst why.

FraudGuard pairs an ML scoring pipeline with an analyst workspace and governance workflows.

**Company:** 3Consulting Limited (Lagos, Nigeria)
**Planned site:** fraudguard.3consult-ng.com (deployment pending verification)